

Periodic Table Fundamentals

Organization of the Periodic Table

Elements in the periodic table are arranged in order of increasing atomic number, with elements in the same vertical column, called a group, sharing similar chemical properties due to having the same number of valence electrons.

Elements in the same horizontal row, called a period, have the same number of electron shells.

Periodic Trends

Atomic radius generally decreases across a period from left to right due to increasing nuclear charge, and increases down a group as additional electron shells are added.

Ionization energy, the energy required to remove an electron from a gaseous atom, generally increases across a period and decreases down a group.

Electronegativity, a measure of an atom's tendency to attract electrons in a bond, generally increases across a period and decreases down a group.